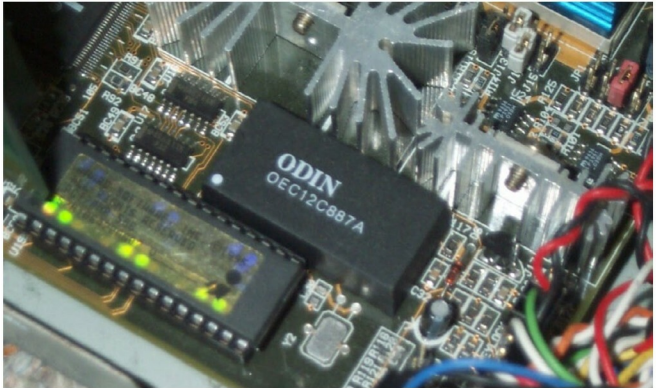


Writing an RTC Driver Based on the SPI Bus

Most computers have one or more hardware clocks that display the current time. These are 'Real Time Clocks' or RTCs. Battery backup is provided for one of these clocks so that time is tracked even when the computer is switched off. RTCs can be used for alarms and other functions like switching computers on or off. This article explains how to write Linux device drivers for SPI-based RTC chips.



We will focus on the RTC DS1347 to explain how device drivers are written for RTC chips. You can refer to the RTC DS1347 datasheet for a complete understanding of this driver.

Linux SPI subsystem

In Linux, the SPI subsystem is designed in such a way that the system running Linux is always an SPI master. The SPI subsystem has three parts, which are listed below.

The SPI master driver: For each SPI bus in the system, there will be an SPI master driver in the kernel, which has routines to read and write on that SPI bus. Each SPI master driver in the kernel is identified by an SPI bus number. For the purposes of this article, let's assume that the SPI master driver is already present in the system.

The SPI slave device: This interface provides a way of describing the SPI slave device connected to the system. In

this case, the slave device is RTC DS1347. Describing the SPI slave device is an independent task that can be done as discussed in the section on 'Registering RTC DS1347 as an SPI slave device'.

The SPI protocol driver: This interface provides methods to read and write the SPI slave device (RTC DS1347).

Writing an SPI protocol driver is described in the section on 'Registering the DS1347 SPI protocol driver'.

The steps for writing an RTC DS1347 driver based on the SPI bus are as follows:

1. Register RTC DS1347 as an SPI slave device with the SPI master driver, based on the SPI bus number to which the SPI slave device is connected.
2. Register the RTC DS1347 SPI protocol driver.
3. Once the probe routine of the protocol driver is called, register the RTC DS1347 protocol driver to read and write routines to the Linux RTC subsystem.